

54.5 - Homework and Discussions

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1 Disc 12

1.1

Let A be any 4×4 matrix, with eigenvalues $5, -3, 2$. Suppose you know that the eigenspace associated to the eigenvalue $\lambda_1 = 5$ is 2-dimensional. Must A be diagonalizable?

Conjecture: for $\lambda_1 = 5$ to be $\in \mathbb{R}^2$, it must be invertible, with $\det(A) \neq 0$, and it requires a_{44} to be $a_{44} - \lambda a_{44} = 0$

If we were to do cofactor expansion, we would find:

$$(5 - \lambda)(-3 - \lambda) \begin{vmatrix} 2 - \lambda & a_{34} \\ a_{43} & a_{44} \end{vmatrix}$$

You can't assume A is diagonal or that the characteristic polynomial factors the way you wrote — that would be assuming the conclusion.

The key insight is that when an eigenvalue has algebraic multiplicity 1, its geometric multiplicity is automatically 1 (you always get at least one eigenvector, and at most as many as the algebraic multiplicity).

Invertibility depends on whether 0 is an eigenvalue — it's not here, so A is invertible, but that's a **separate fact and irrelevant to diagonalizability**.

How do you check if 0 is an eigen value... **Independence properties:**

- If $\det(A) = 0$ then 0 is an eigenvalue.
- Check for linear independence.
- In this problem... you don't need to compute anything. The mere fact that 0 isn't an eigenvalue is sufficient.

We are told that $\lambda_1 = 5$ is 2d.

The logic is that the algebraic multiplicities must sum to exactly 4, because we are in $\text{rank}(4)$ and we have the condition that $\det(A) \neq 0$.

This implies that our characteristic polynomial is:

$$p(\lambda) = (5 - \lambda)^2(-3 - \lambda)(2 - \lambda)$$

1.2

- a. The trace of a square matrix A is the sum of its diagonal elements and is denoted by $Tr(A)$. It can be verified with $Tr(AB) = Tr(BA)$. In fact, trace is cyclic, $Tr(ABC) = Tr(CAB) = Tr(BCA)$. Show that if $A \wedge B$ are similar, then $Tr(A) = Tr(B)$

We want to show that $A \wedge B$ have the condition that $det(A) \neq 0 \wedge det(B) \neq 0 \wedge A \sim B \implies A = PBP^{-1}$:

$$\begin{aligned} A &= B \\ &= PBP^{-1} \\ &= PP^{-1}B \end{aligned}$$

(Same logic could be applied for A) $= IB$
 $A = B$

We can move the multiplicative properties of PBP^{-1} because of we know $Tr(ABC)$ is cyclic. now that we showed $Tr(A) = Tr(B)$, this implies that the sum of its diagonal elements are equal

- b. Prove that if A is diagonalizable, then the trace is just the sum of the eigenvalues of A .

Let $M_c \in \mathbb{C}^n$, if $det(A) \neq 0$, then we can safely compute the characteristic eigenvalues for A diagonal. $A = PDP^{-1} \rightarrow A = PP^{-1}D \rightarrow A = ID \rightarrow A = D$, where P is another matrix that is equivalent to A , and D is the diagonal.

If A is diagonalizable, then $Tr(A) = Tr(PDP^{-1}) \rightarrow Tr(PP^{-1}D) \implies Tr(ID) \rightarrow Tr(A) = Tr(D)$. Thus the values of $Tr(D)$ on the diagonal are A 's eigen values and therefore the trace is just the sum of the eigenvalues of A .

1.3

<https://www.pinecone.io/learn/vector-similarity/> Let A be a real symmetric matrix (all entries are real and $A^T = A$). Define $q = \bar{x}^T Ax$. Show that q is real (i.e. $\bar{q} = q$). Then show that if $Ax = \lambda x$, then λ must be real and the real part of x is also an eigenvector of A .

To show $\bar{q}$ is real: This needs to be fixed, you should find that you get back to the start, thus its cyclic and they are real

$$\begin{aligned} \bar{q} &= \bar{x}^T Ax \\ q &= x^T \bar{A} \bar{x} \\ q &= \bar{q} \end{aligned}$$

To show that $Ax = \lambda x$

$$\begin{aligned} Ax &= \lambda x \\ \bar{x}^T Ax &= \bar{x}^T \lambda x \quad (\text{multiply both sides on the left by } \bar{x}^T) \\ \bar{x}^T Ax &= \lambda \bar{x}^T x \quad (\text{notice the identity } \bar{q}) \\ \frac{\bar{x}^T Ax}{\bar{x}^T x} &= \lambda \end{aligned}$$

Here you need to show that $\bar{x}^T x \rightarrow \sum_{k=0} |k_k|^2$ is a real number. Thus we conclude that λ is real.

To show that x is an eigenvector of A , we use $Re(z), Im(z)$. Let $x = u - vi$

$$\begin{aligned} Ax &= \lambda x \\ A(u - vi) &= \lambda(u - vi) \\ Au - vAi &= \lambda u - v\lambda i \end{aligned}$$

We can see the real part of the equation that x 's eigen vector is apart of. I.e. $Au \wedge \lambda u$. Whenever we have something like $Au - vAi = \lambda u - v\lambda i$ we implicitly are showing two equations, i.e.:

$$\begin{aligned} Au - vAi &= \lambda u - v\lambda i \\ (Au + \lambda u) &= (vAi - v\lambda i) \end{aligned}$$

1.4 HW 5.5

5. Let each matrix act on $\mathbb{C}^2$. Find eigenvalues and a basis for each eigenspace $\in \mathbb{C}^2$.

$$\begin{bmatrix} 0 & 1 \\ -8 & 4 \end{bmatrix}$$

Steps:

1. Find Characteristic equation, call it q
2. Find $Re(z) \wedge Im(z)$ with $quadraticFormula(q) = z$
3. Substitute z in A , i.e. $(A - \lambda I) = 0$
4. Find $\mathbf{v}_1$ by expressing in terms of row 1, setting the free variable equal to 1
5. With $\mathbf{v}_1$, $conjugate(\mathbf{v}_1)$ to find $\mathbf{v}_2$, i.e. conjugate for free

$$\begin{bmatrix} 0 - \lambda & 1 \\ -8 & 4 - \lambda \end{bmatrix}$$

$$\rightarrow (0 - \lambda)(4 - \lambda) = \lambda^2 - 4\lambda + 8 \rightarrow \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(8)}}{2(1)}$$

$$z = 2 \pm 2i \rightarrow A - zI \rightarrow \begin{bmatrix} 0 - (2 + 2i) & 1 \\ -8 & 4 - (2 + 2i)s \end{bmatrix}$$

$$\begin{bmatrix} -2 - 2i & 1 \\ -8 & 2 + 2is \end{bmatrix}$$

$$\rightarrow (-2 - 2i)x_1 + x_2 = x_2 = (2 + 2i)x_1$$

Key idea is we can express it in any term, and in this case we chose x_2 because it was the easiest to find (i.e. x_2 already had the coefficient of 1, thus it is trivially solved. Finally, we choose the free variable $x_1 = 1$ and express it as:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 + 2i \end{bmatrix}$$

7. List eigenvalues of A . $\mathbf{x} \mapsto A\mathbf{x}$ is the composition of a rotation and a scaling. Give the angle φ of the rotation, where $-\pi < \varphi \leq \pi$, and give the scale factor r .

$$\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$

Steps:

1. find $r = |\lambda| = \sqrt{a^2 + b^2}$
 2. Set up and solve $V = r \begin{bmatrix} \frac{a}{r} & -\frac{b}{r} \\ \frac{b}{r} & \frac{a}{r} \end{bmatrix}$
 3. Find $\det(V)$, which entails taking $(V - \lambda I) = 0 \rightarrow z = \lambda^2 - 2\lambda\sqrt{3} + 4$.
 4. Calling our found polynomial z , apply `quadraticEquation(z)` to get $\sqrt{3} \pm i$
 5. Find the angle you rotate by with $\arctan(b/a)$ taken from our found V .
13. Find an invertible matrix P and a matrix of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ s.t.

the given matrix has form $A = PCP^{-1}$ for $A = \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1-\lambda & -2 \\ 1 & 3-\lambda \end{bmatrix} \rightarrow (1-\lambda)(3-\lambda) - (-2)(1) \rightarrow \lambda^2 - 4\lambda + 5$$

$$\frac{-(-4) \pm \sqrt{4^2 - 4(1)(5)}}{2} = \frac{4 \pm \sqrt{16-20}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4}{2} \pm \frac{\sqrt{-4}}{2} = 2 + \frac{2i}{2} = 2 \pm i$$

$$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1-(2 \pm i) & -2 \\ 1 & 3-(2 \pm i) \end{bmatrix} \rightarrow \begin{bmatrix} -1-i & -2 \\ 1 & 1-i \end{bmatrix} \rightarrow \mathbf{x}_2 = -1 + i$$

$$\mathbf{x}_2 \begin{bmatrix} -1+i \\ 1 \end{bmatrix} \Rightarrow \mathbf{v}_1 = \begin{bmatrix} -1+i \\ 1 \end{bmatrix} \text{ i.e. we found the basis, now we find } C = P^{-1}AP$$

$$P = [Re(z) \ Im(z)] \rightarrow P = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}, \ P^{-1} = \frac{1}{(-1)(0) - (1)(1)} \begin{bmatrix} 0 & -1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

Now with $P^{-1}AP$ found, we can finally do the following to find C :

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$$

What I had to learn the hard way was scalar multiplication and addition such that we preserve the columns of the matrices. Observe:

$$\begin{bmatrix} (0)(1) + (1)(1) & (0)(-2) + (1)(3) \\ (1)(1) + (1)(1) & (1)(-2) + (1)(3) \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} (1)(-1) + (3)(1) & (1)(1) + (3)(0) \\ (2)(-1) + (1)(1) & (2)(1) + (1)(0) \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} = C$$

It should be noted that we can use $-1 \pm i$, and either way we still get a correct answer. Something you got tripped up on, i.e. the textbook solution uses

$$\mathbf{v} = \begin{bmatrix} -1-i \\ 1 \end{bmatrix}$$

21. From example 2 in the textbook, solve the first equation in (2) for x_2 in terms of x_1 , and from that produce the eigenvector $\mathbf{y} = \begin{bmatrix} 2 \\ -1 + 2i \end{bmatrix}$ for the matrix A . Show that this $\mathbf{y}$ is a complex multiple of the vector $\mathbf{v}_1$ used in example 2.

$$(-.3 + .6i)x_1 - .6x_2 = 0, (.75)x_1 + (3 + .6i)x_2 = 0, \mathbf{v}_1 = \begin{bmatrix} -2 - 4i \\ 5 \end{bmatrix}$$

$$(\frac{-.3 + 6i}{.6})x_1 + x_2 = 0$$

$$(\frac{-1 + 2i}{2})x_1 = x_2$$

$$(-2 + i)x_1 = x_2$$

$$x_2 = -1 + 2i$$

✓

$$(\frac{-1 + 2i}{2})x_1 = x_2$$

$$(-\frac{1}{2} + i)x_1 = x_2$$

$$(-\frac{1}{2} + i)(2) = x_2$$

$$-1 + 2i = x_2$$

- Note to self: you messed up on multiple times. When you find the scalar c as $\frac{-1+2i}{2}$, you are allowed to simplify it more. HOW STUPID, simply do it correctly.
- Behind the scenes, understand that when we choose x_1 to be 2, that we get $\mathbf{y}$

To show $\mathbf{y}$ is a complex multiple of the vector $\mathbf{v}_1$, we find a scalar $\alpha \in \mathbb{C}$ s.t. $\mathbf{y} = \alpha \mathbf{v}_1$, i.e. we solve the system $y = \alpha v_{1,1} \wedge \alpha v_{1,2} = y_2$:

$$\begin{bmatrix} 2 \\ -1 + 2i \end{bmatrix} \begin{bmatrix} -2 - 4i \\ 5 \end{bmatrix} \rightarrow \alpha = \frac{y_1}{v_{1,1}} = \frac{2}{-2 - 4i} = \frac{2}{-2 - 4i} \cdot \frac{-2 + 4i}{-2 + 4i}$$

We apply difference of squares formula to find

$$\frac{-4 + 8i}{20} = \frac{-1}{5} + \frac{2}{5}i = y = \alpha v_1$$

- We rationalize $\frac{y_1}{v_{1,1}} = \frac{2}{-2-4i}$ because we don't like imaginary numbers in the denominator.

1.5 Discussion 15

Problem 1:

- $p \in \text{Row}(A)$ and $u \in \text{Null}(A)$
- $\text{rank}A + \text{nullity}A = \# \text{columns} \in A$

Problem 2:

- Why do they keep mentioning $Rx = 0$?
- The columns of A are linearly independent, thus it follows that $A = QR$, where R is $n \times n$, with A as $m \times n$, R must be invertible ?

1.6 Quiz Algorithm

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 4 & 6 \end{bmatrix}$$

Show with row sum that we have the identity $\lambda(\text{row1})$ i.e. $6(r1)$

Algorithm:

- $\text{ref}(C|B)$ s.t. C becomes I to find $P_{C \leftarrow B}$
- Find $P_{C \leftarrow B}^{-1}$ with the inverse formula